Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Why might a linear Kalman filter become unstable? Select all that apply.

1 / 1 point

- [ ] A linear Kalman filter will become unstable if either the process noise w_k or sensor noise v_k is very large.
- [ ] A linear Kalman filter will become unstable if the system whose state it is estimating is unstable.
- [x] A linear Kalman filter will become unstable if $\Sigma^+_{\tilde{x},k}$ loses symmetry.

Correct
Yes, this is true.

- [ ] A linear Kalman filter will become unstable if its computation of $\hat{x}^+_k$ is not equal to the true x_k.
- [x] A linear Kalman filter will become unstable if $\Sigma^+_{\tilde{x},k}$ becomes non-positive-definite (has nonpositive eigenvalues).

Correct
Yes, this is true.

2. Which of the following matrices are positive-definite (i.e., are both symmetric and have strictly positive eigenvalues)? You may use Octave to compute the eigenvalues. Select all that apply.

1 / 1 point

- [ ] $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$
- [x] $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

Correct
Yes. This matrix is positive definite. All of its eigenvalues (0.48, 2, 3.41) are positive and the matrix is symmetric.

- [x] $\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix}$

Correct
Yes. This matrix is positive definite. All of its eigenvalues (0.18, 21.82) are positive and the matrix is symmetric.

- [ ] $[-1]$
- [ ] $\begin{bmatrix} 2 & 6 \\ 6 & 18 \end{bmatrix}$

3. In this question, you will use the Higham method to convert a nonpositive-definite matrix to the "nearest" positive-semidefinite matrix. You will need to compute a singular-value decomposition, which can be done in Octave using the command: `[U,S,V] = svd(Sigma);`

1 / 1 point

What is the "nearest" positive-semidefinite matrix to the 1x1 matrix $\Sigma = -1$? Enter your answer with one digit to the right of the decimal point.

0

Correct
Yes. That is the correct answer.